

Lesson 10: Data Preparation and Transformation using Tableau Prep

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What you will learn ?

- a) Connect and load data into Tableau Prep
- b) Clean and profile data
- c) Group and standardize values
- d) Merge mismatched fields
- e) Perform aggregations in Prep
- f) Apply filters and transformations
- g) Create and automate data preparation flows

You need Tableau Prep Builder for this Lesson. You need to download and install. Here are the steps :

1. Download

- Download the Windows installer (.exe) directly from the official [Tableau Prep Builder Releases Page](#) (or use the [Tableau Prep Free Trial Page](#)).

2. Install

- Open your **Downloads** folder and run the .exe file.
- Accept the license agreement, click **Install**, and confirm **Yes** on the Windows prompt.

3. Activate & Launch

- Open **Tableau Prep Builder** from the Start Menu.
- Select **Activate with a product key** (enter your Tableau Desktop/Student key) or **Start a Free Trial**.

Dataset: Custom Designed Students Dataset [csv files]

You can also create database using Python code. Here is the code

```
import pandas as pd
import numpy as np

# We need custom CSV datasets specifically designed for Tableau
# Prep features in Practical 10:
# a) Connect & load data
# b) Clean and profile data (data type issues, extra spaces,
# nulls, outliers)
# c) Group and standardize values (pronunciation/spelling
# variations, e.g. "Comp Sci", "CS", "Computer Science", "CSE",
# "Electronic Eng", "ECE")
# d) Merge mismatched fields (e.g., across two unioned files
# or separate columns: "Maths" vs "Mathematics", "Std_ID" vs
# "Student_ID")
# e) Perform aggregations in Prep (e.g., Aggregate step: Total
# Score, Subject Average, Group by Student/Term)
# f) Apply filters and transformations (e.g., Filter out
# dropped students, calculate weighted GPA / Grade, split
# columns)
```

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```
# g) Create and automate data preparation flows (Output step
to .hyper / .csv)

# Table 1: Raw Student Assessment Log (Term 1) ->
Student_Assessments_Term1.csv
# Contains messy department names for (c) Group & Standardize,
inconsistent casings, text dates, extra spaces

term1_data = {
    'Student_ID': ['S101', 'S101', 'S102', 'S102', 'S103',
'S103', 'S104', 'S104', 'S105', 'S105', 'S106', 'S106',
'S107', 'S107', 'S108', 'S108'],
    'Student_Name': ['Aarav Sharma ', 'Aarav Sharma', 'priya
VERMA', 'priya VERMA', 'Rohan Gupta', 'Rohan Gupta', 'Sneha
Patel', 'Sneha Patel', 'aditya Rao ', 'aditya Rao ', 'Ananya
Singh', 'Ananya Singh', 'Rahul Nair', 'Rahul Nair', 'Kavya
Reddy', 'Kavya Reddy'],
    'Department': ['Computer Science', 'Comp Sci',
'Electronics', 'ECE', 'CS', 'CSE', 'Information Tech', 'IT',
'Mechanical', 'Mech Engg', 'Computer Science', 'CS', 'ECE',
'Electronics Engg', 'IT', 'Info Tech'],
    'Assessment_Type': ['Midterm', 'Final', 'Midterm',
'Final', 'Midterm', 'Final', 'Midterm', 'Final', 'Midterm',
'Final', 'Midterm', 'Final', 'Midterm', 'Final', 'Midterm',
'Final'],
    'Exam_Date': ['2025/09/15', '2025/11/20', '15-09-2025',
'20-11-2025', '2025.09.15', '2025.11.20', '2025/09/15',
'2025/11/20', '2025/09/15', '2025/11/20', '2025/09/15',
'2025/11/20', '2025/09/15', '2025/11/20', '2025/09/15',
'2025/11/20'],
    'Subject_Code': ['CS101', 'CS101', 'EC101', 'EC101',
'CS101', 'CS101', 'IT101', 'IT101', 'ME101', 'ME101', 'CS101',
'CS101', 'EC101', 'EC101', 'IT101', 'IT101'],
    'Marks_Obtained': [85, 92, 70, 78, 45, 52, 90, 96, 35, 40,
88, 94, 65, 72, 92, 98],
    'Max_Marks': [100, 100, 100, 100, 100, 100, 100, 100, 100,
100, 100, 100, 100, 100, 100, 100],
    'Enrollment_Status': ['Active', 'Active', 'Active',
'Active', 'Active', 'Active', 'Dropped',
'Dropped', 'Active', 'Active', 'Active', 'Active', 'Active',
'Active']
}

# Table 2: Student Assessment Log (Term 2) with Mismatched
Column Headers -> Student_Assessments_Term2.csv
# For (d) Merge Mismatched Fields (e.g. Std_ID instead of
Student_ID, Score instead of Marks_Obtained, Dept instead of
Department)

term2_data = {
    'Std_ID': ['S101', 'S101', 'S102', 'S102', 'S103', 'S103',
'S104', 'S104', 'S106', 'S106', 'S107', 'S107', 'S108',
'S108'],
```

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```

'Full_Name': ['Aarav Sharma', 'Aarav Sharma', 'Priya
Verma', 'Priya Verma', 'Rohan Gupta', 'Rohan Gupta', 'Sneha
Patel', 'Sneha Patel', 'Ananya Singh', 'Ananya Singh', 'Rahul
Nair', 'Rahul Nair', 'Kavya Reddy', 'Kavya Reddy'],
'Dept': ['Computer Science', 'CS', 'ECE', 'Electronics',
'CSE', 'CS', 'IT', 'Information Tech', 'CSE', 'Computer
Science', 'ECE', 'Electronics', 'IT', 'IT'],
'Exam_Type': ['Midterm', 'Final', 'Midterm', 'Final',
'Midterm', 'Final', 'Midterm', 'Final', 'Midterm', 'Final',
'Midterm', 'Final', 'Midterm', 'Final'],
'Assessment_Date': ['2026/02/10', '2026/04/25',
'2026/02/10', '2026/04/25', '2026/02/10', '2026/04/25',
'2026/02/10', '2026/04/25', '2026/02/10', '2026/04/25',
'2026/02/10', '2026/04/25', '2026/02/10', '2026/04/25'],
'Course_Code': ['CS201', 'CS201', 'EC201', 'EC201',
'CS201', 'CS201', 'IT201', 'IT201', 'CS201', 'CS201', 'EC201',
'EC201', 'IT201', 'IT201'],
'Score': [88, 95, 75, 82, 50, 60, 94, 98, 90, 96, 70, 75,
95, 100],
'Total_Possible': [100, 100, 100, 100, 100, 100, 100, 100, 100, 100,
100, 100, 100, 100],
'Enrollment_Status': ['Active', 'Active', 'Active',
'Active', 'Active', 'Active', 'Active', 'Active',
'Active', 'Active', 'Active', 'Active', 'Active']
}

# Table 3: Course Credit & Grading Benchmark ->
Course_Credit_Master.csv

course_data = {
    'Subject_Code': ['CS101', 'CS201', 'EC101', 'EC201',
'IT101', 'IT201', 'ME101'],
    'Subject_Title': ['Data Structures', 'Algorithms',
'Digital Logic', 'Signals & Systems', 'Database Systems',
'Cloud Computing', 'Thermodynamics'],
    'Credit_Hours': [4, 4, 3, 4, 4, 4, 3],
    'Passing_Marks': [40, 40, 40, 40, 40, 40, 40]
}

df_t1 = pd.DataFrame(term1_data)
df_t2 = pd.DataFrame(term2_data)
df_c = pd.DataFrame(course_data)

df_t1.to_csv('Student_Assessments_Term1.csv', index=False)
df_t2.to_csv('Student_Assessments_Term2.csv', index=False)
df_c.to_csv('Course_Credit_Master.csv', index=False)

print("All 3 CSV datasets for Lesson 10 created successfully.")

```

Here is the complete schema design and CSV datasets tailored specifically for Tableau Prep Builder to fulfill all activities of Practical 10 (a through g).

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Dataset Architecture and Schema Design

To practice every feature of Tableau Prep (profiling, grouping, merging mismatched union columns, aggregations, calculated filters, and automated flow generation), we use 3 specialized CSV files:

1. Student_Assessments_Term1.csv (Term 1 Assessment Log)
 - Purpose: Data profiling, handling extra spaces, mixed casings, inconsistent date text strings, and spelling variations in departments.
 - Columns: Student_ID, Student_Name, Department, Assessment_Type, Exam_Date, Subject_Code, Marks_Obtained, Max_Marks, Enrollment_Status.
2. Student_Assessments_Term2.csv (Term 2 Assessment Log with Mismatched Schemas)
 - Purpose: Practicing Union and resolving mismatched field names (e.g., Std_ID vs Student_ID, Score vs Marks_Obtained, Course_Code vs Subject_Code).
 - Columns: Std_ID, Full_Name, Dept, Exam_Type, Assessment_Date, Course_Code, Score, Total_Possible, Enrollment_Status.
3. Course_Credit_Master.csv (Subject Metadata & Weightages)
 - Purpose: Prep Joins, weighted GPA calculations, credit hours, and subject categorization.
 - Columns: Subject_Code, Subject_Title, Credit_Hours, Passing_Marks.

Mapping to Practical 10 Objectives

Sub-task	Activity	Practical Execution in Tableau Prep
a	Connect and load data	Connect to text files (Term1, Term2, and Course_Master) in Tableau Prep.
b	Clean and profile data	Use Prep's Profile pane to inspect value distributions, summary cards, and detect data type errors.
c	Group and standardize values	Use Group Values (Pronunciation, Common Characters, or Manual) to unify "Comp Sci", "CS", "CSE" --> "Computer Science".
d	Merge mismatched fields	Union Term 1 and Term 2, then drag and drop mismatched columns (Std_ID onto Student_ID, Score onto Marks_Obtained) to merge them.
e	Perform aggregations in Prep	Add an Aggregate Step to compute student total score, overall average, and exam count grouped by Student_ID & Term.

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Sub-task	Activity	Practical Execution in Tableau Prep
f	Apply filters and transformations	Filter out "Dropped" enrollment records, create calculated fields for Percentage, Grade, and Status flags.
g	Create and automate data flows	Build an Output Step to export the cleaned, unified pipeline to a .hyper extract or .csv file.

Raw CSV Data (Create these 3 files on your computer). You can create different csv files using Notepad/text Editor. Copy paste the data in Notepad/text Editor and save the files with .csv extension.

File 1: Student_Assessments_Term1.csv

```
Student_ID,Student_Name,Department,Assessment_Type,Exam_Date,
Subject_Code,Marks_Obtained,Max_Marks,Enrollment_Status
S101, Aarav Sharma ,Computer
Science,Midterm,2025/09/15,CS101,85,100,Active
S101,Aarav Sharma,Comp
Sci,Final,2025/11/20,CS101,92,100,Active
S102,priya VERMA,Electronics,Midterm,15-09-
2025,EC101,70,100,Active
S102,priya VERMA,ECE,Final,20-11-2025,EC101,78,100,Active
S103,Rohan Gupta,CS,Midterm,2025.09.15,CS101,45,100,Active
S103,Rohan Gupta,CSE,Final,2025.11.20,CS101,52,100,Active
S104,Sneha Patel,Information
Tech,Midterm,2025/09/15,IT101,90,100,Active
S104,Sneha Patel,IT,Final,2025/11/20,IT101,96,100,Active
S105,aditya Rao
,Mechanical,Midterm,2025/09/15,ME101,35,100,Dropped
S105,aditya Rao ,Mech
Engg,Final,2025/11/20,ME101,40,100,Dropped
S106,Ananya Singh,Computer
Science,Midterm,2025/09/15,CS101,88,100,Active
S106,Ananya Singh,CS,Final,2025/11/20,CS101,94,100,Active
S107,Rahul Nair,ECE,Midterm,2025/09/15,EC101,65,100,Active
S107,Rahul Nair,Electronics
Engg,Final,2025/11/20,EC101,72,100,Active
S108,Kavya Reddy,IT,Midterm,2025/09/15,IT101,92,100,Active
S108,Kavya Reddy,Info
Tech,Final,2025/11/20,IT101,98,100,Active
```

File 2: Student_Assessments_Term2.csv

```
Std_ID,Full_Name,Dept,Exam_Type,Assessment_Date,Course_Code,S
core>Total_Possible,Enrollment_Status
S101,Aarav Sharma,Computer
Science,Midterm,2026/02/10,CS201,88,100,Active
S101,Aarav Sharma,CS,Final,2026/04/25,CS201,95,100,Active
S102,Priya Verma,ECE,Midterm,2026/02/10,EC201,75,100,Active
```

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```
S102,Priya
Verma,Electronics,Final,2026/04/25,EC201,82,100,Active
S103,Rohan Gupta,CSE,Midterm,2026/02/10,CS201,50,100,Active
S103,Rohan Gupta,CS,Final,2026/04/25,CS201,60,100,Active
S104,Sneha Patel,IT,Midterm,2026/02/10,IT201,94,100,Active
S104,Sneha Patel,Information
Tech,Final,2026/04/25,IT201,98,100,Active
S106,Ananya Singh,CSE,Midterm,2026/02/10,CS201,90,100,Active
S106,Ananya Singh,Computer
Science,Final,2026/04/25,CS201,96,100,Active
S107,Rahul Nair,ECE,Midterm,2026/02/10,EC201,70,100,Active
S107,Rahul
Nair,Electronics,Final,2026/04/25,EC201,75,100,Active
S108,Kavya Reddy,IT,Midterm,2026/02/10,IT201,95,100,Active
S108,Kavya Reddy,IT,Final,2026/04/25,IT201,100,100,Active
```

File 3: Course_Credit_Master.csv

```
Subject_Code,Subject_Title,Credit_Hours,Passing_Marks
CS101,Data Structures,4,40
CS201,Algorithms,4,40
EC101,Digital Logic,3,40
EC201,Signals & Systems,4,40
IT101,Database Systems,4,40
IT201,Cloud Computing,4,40
ME101,Thermodynamics,3,40
```

A: Connect and Load Data into Tableau Prep

Objective: Establish connections to the local CSV data sources (Student_Assessments_Term1.csv, Student_Assessments_Term2.csv, and Course_Credit_Master.csv) and initiate a visual data preparation workflow.

Tableau Prep Builder utilizes an input step pipeline architecture. Connecting multiple disparate files into a unified workspace establishes the foundation for schema validation, data type detection, and downstream data profiling.

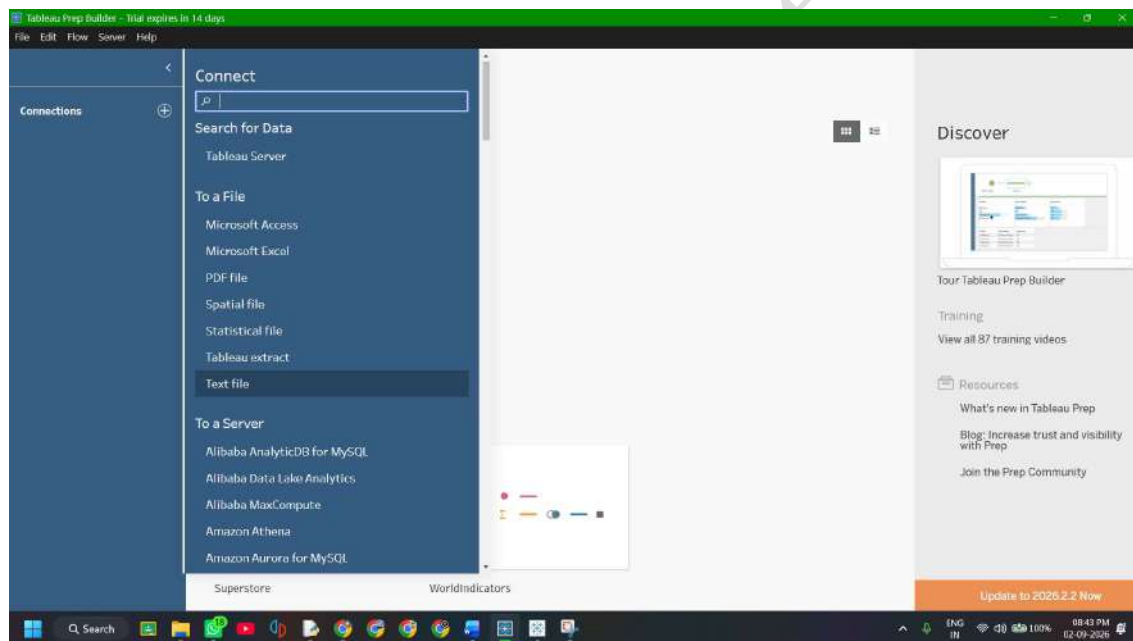
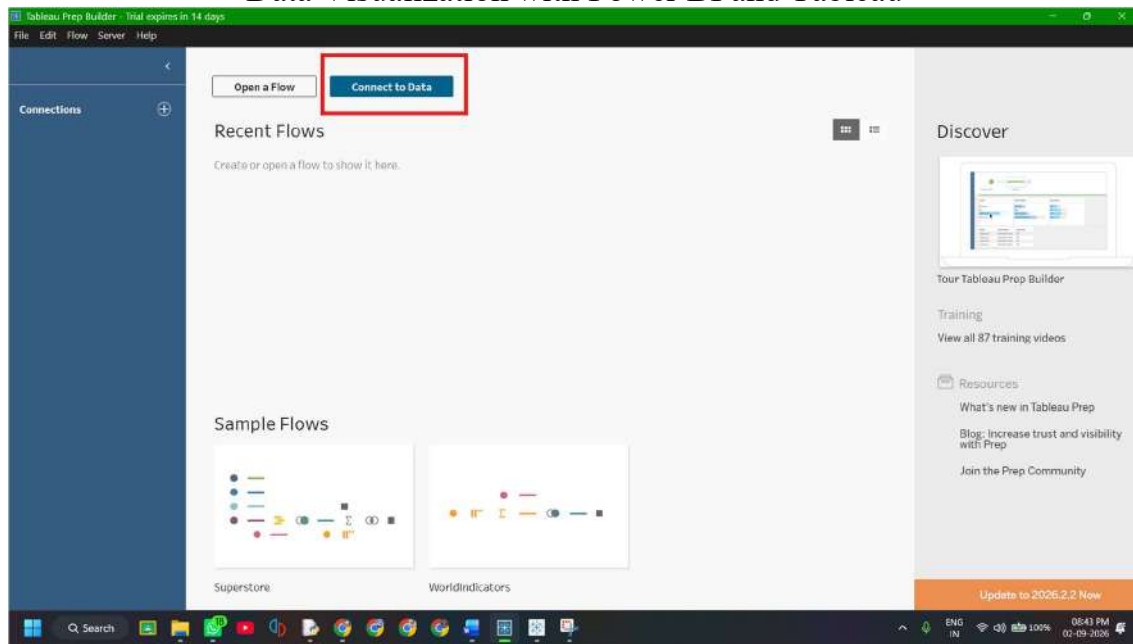
Step 1: Open Tableau Prep Builder

1. Launch **Tableau Prep Builder** on your machine.
2. On the start screen, locate the **Connections** pane on the left-hand side.

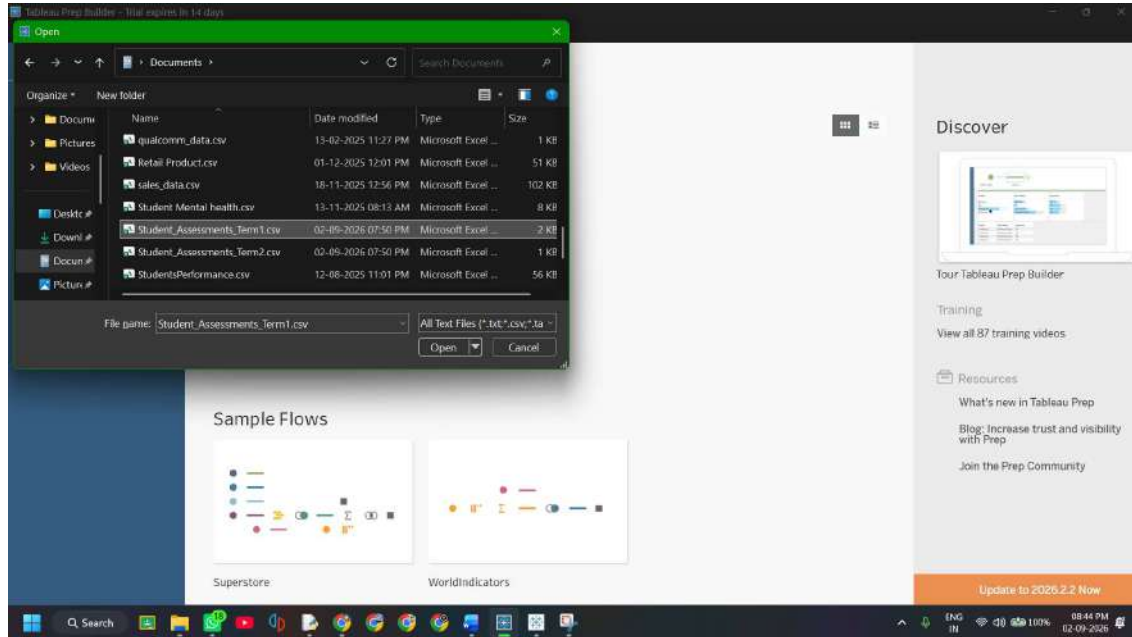
Step 2: Load the First Dataset (Student_Assessments_Term1.csv)

1. In the **Connections** pane, click the + (**Add connection**) button.

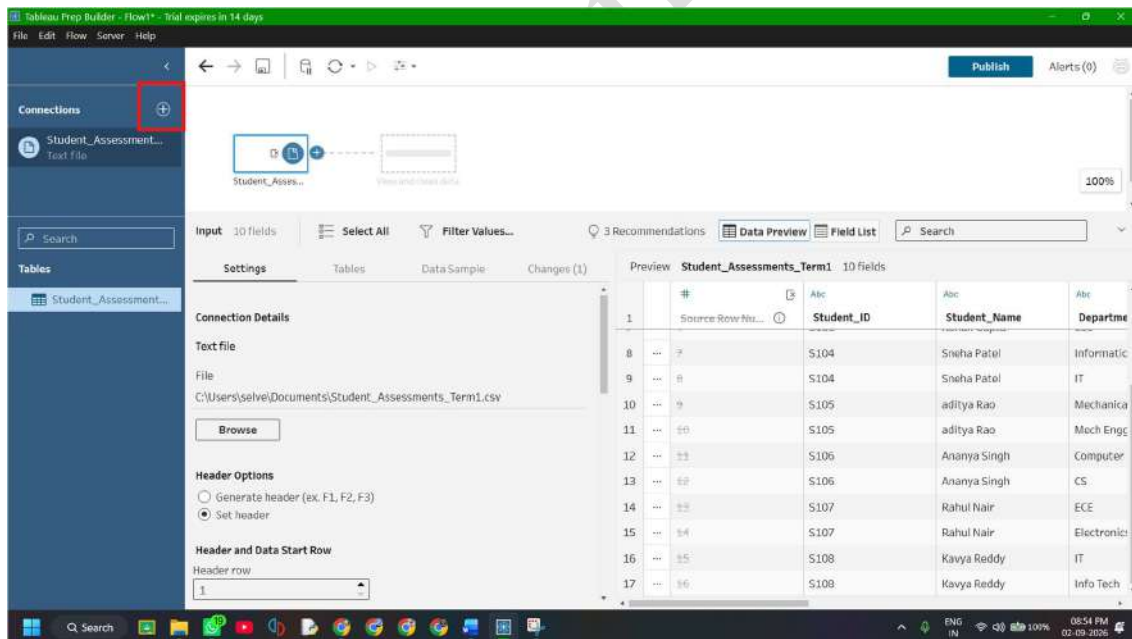
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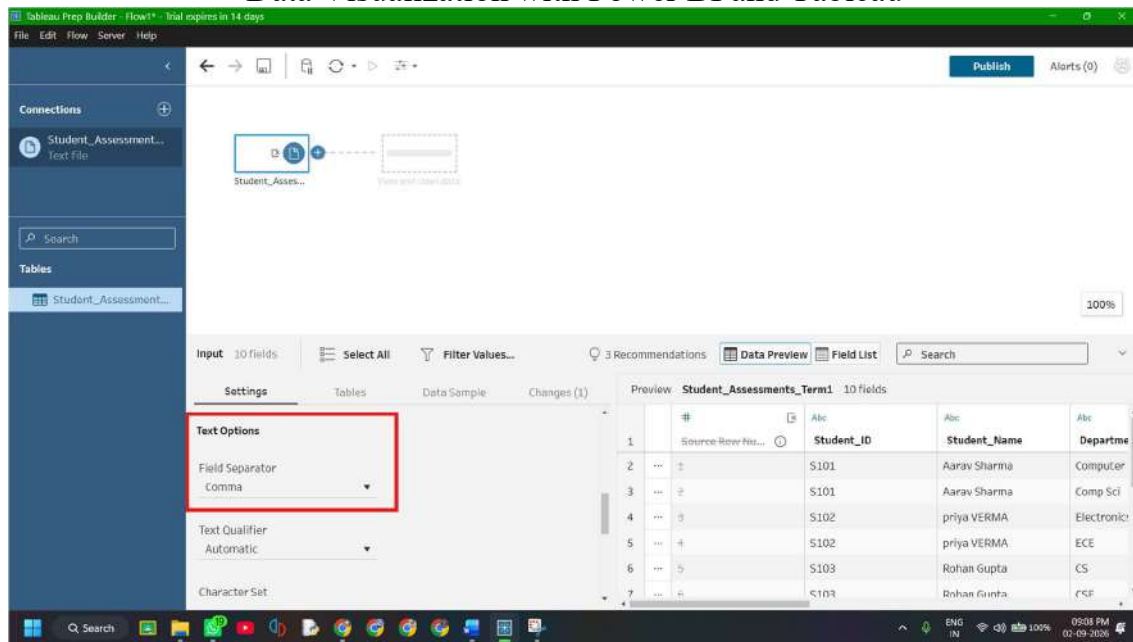


2. In the list of file formats, select **Text file**.
3. Browse to the folder where you saved the CSVs, select **Student_Assessments_Term1.csv**, and click **Open**.
4. An **Input Step icon** named Student_Assessments_Term1 will appear on the Flow canvas.



5. In the bottom **Input configuration** pane, verify:
 - Field Separator: Comma (,)

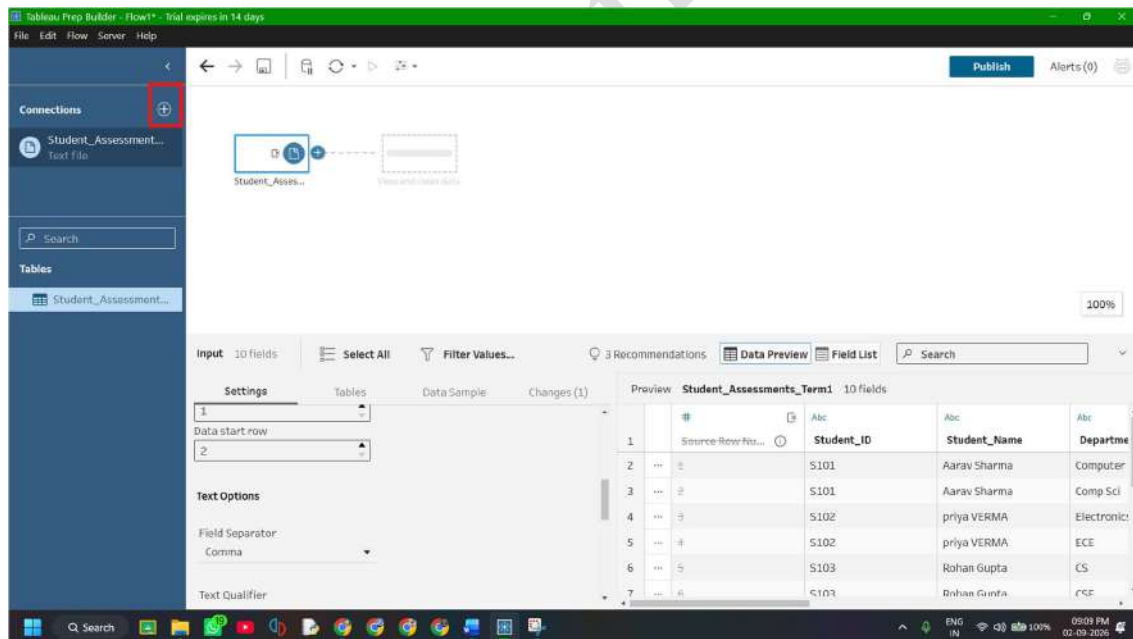
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- **First row contains header:** Checked

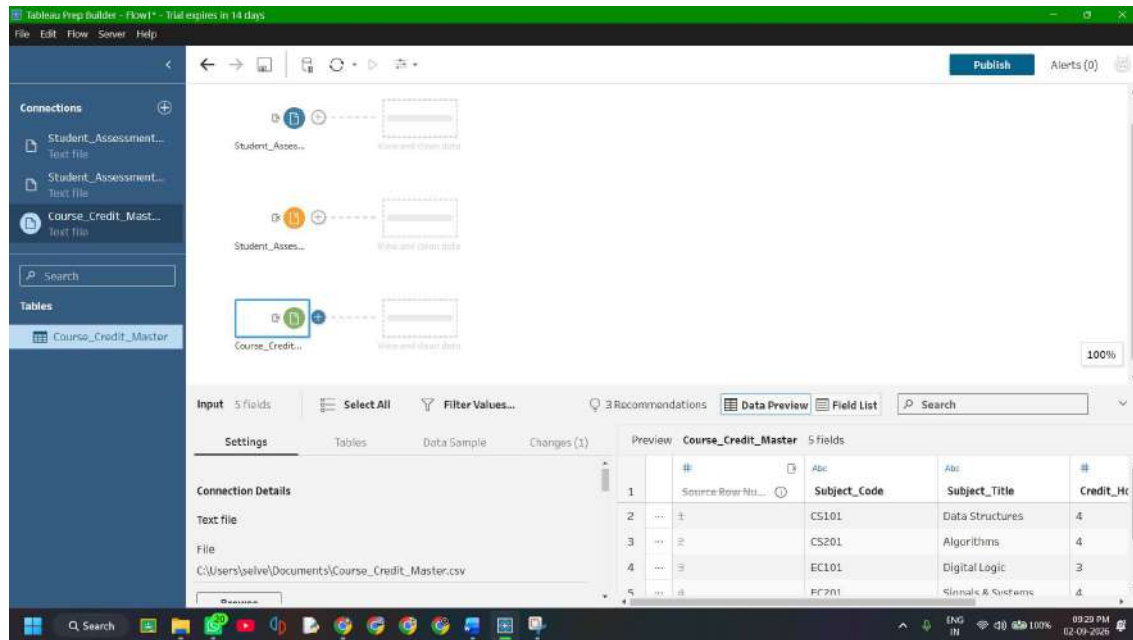
Step 3: Load the Second and Third Datasets

1. In the top-left **Connections** pane, click the + (**Add connection**) button again.



2. Select **Text file**, browse and select **Student_Assessments_Term2.csv**, and click **Open**.
3. Click + (**Add connection**) once more → select **Text file** → choose **Course_Credit_Master.csv** → click **Open**.

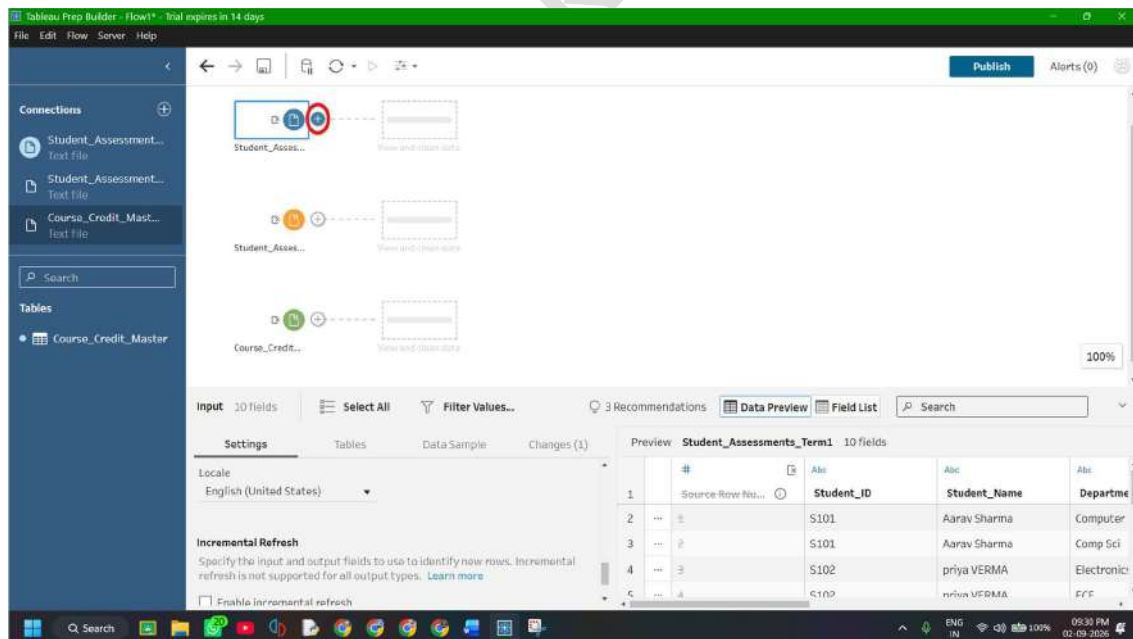
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4. You will now see **three separate Input** icons on your Flow canvas.

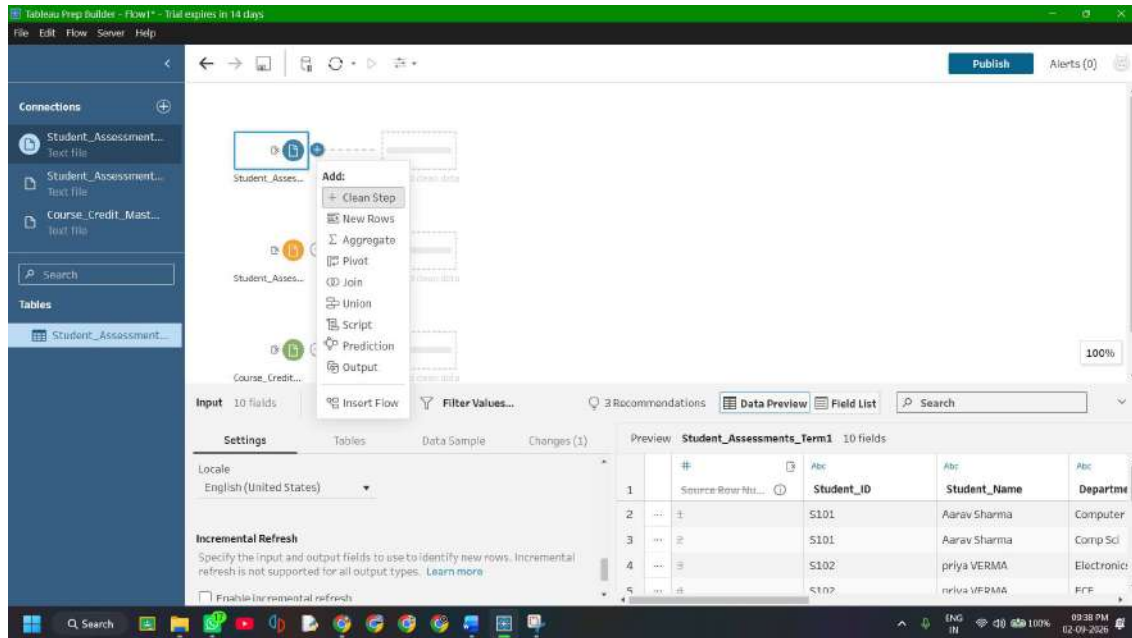
Step 4: Add an Initial Clean Step to Term 1

1. On the Flow canvas, hover your mouse over the **Student_Assessments_Term1** input icon.
2. Click the small + icon that appears to its right.

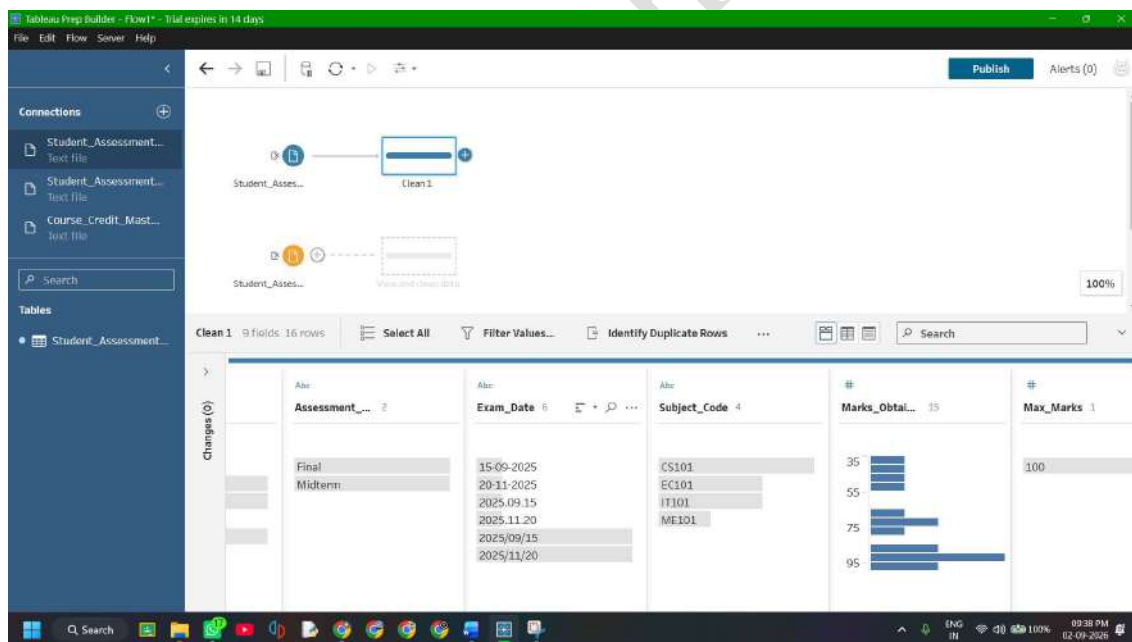


Data Visualization with Power BI and Tableau

3. Select **Clean Step** (the step with the cleaning tool icon).



4. The **Profile Pane** will open at the bottom, displaying interactive profile cards for each of the 9 columns in Term 1.



B: Clean and Profile Data

Objective: Use Tableau Prep Builder's interactive **Profile Pane** to identify data anomalies, whitespace issues, date formatting discrepancies, and inconsistent casing, then apply standard cleaning operations.

Unlike traditional spreadsheets, Tableau Prep provides real-time profiling cards that visualize distributions, frequency counts, and metadata data types for every field. Profiling allows you to quickly detect hidden formatting defects (such as extra spaces in names like ' Aarav

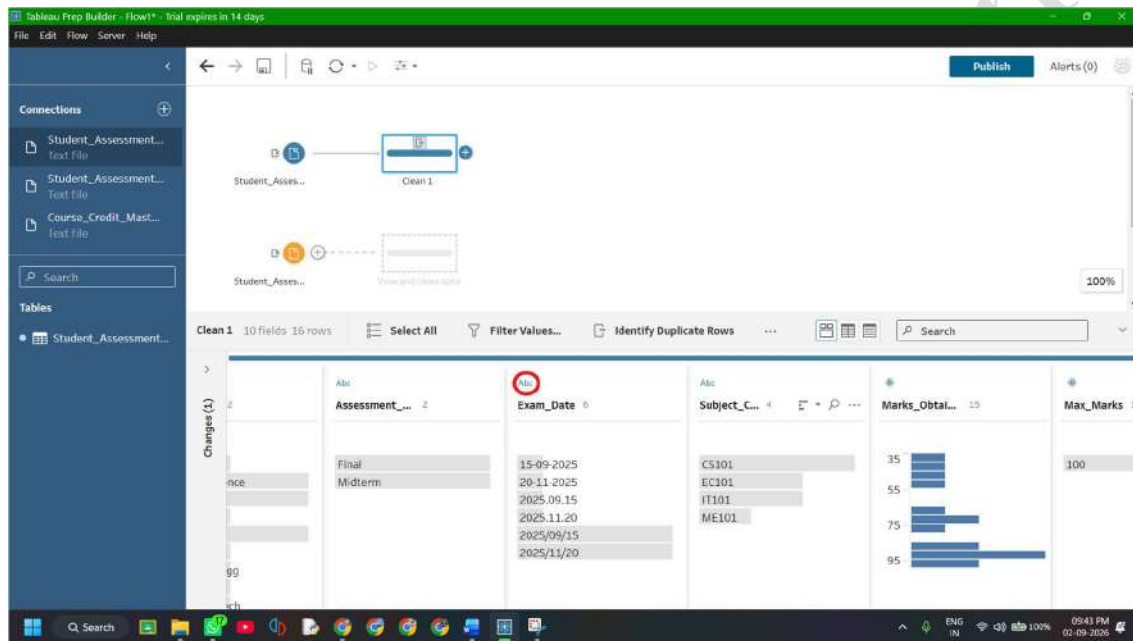
Dr. Mahendra K.

Data Visualization with Power BI and Tableau

Sharma ' or non-standard date strings) before standardizing categorical groups or performing joins.

Step 1: Inspect the Profile Pane

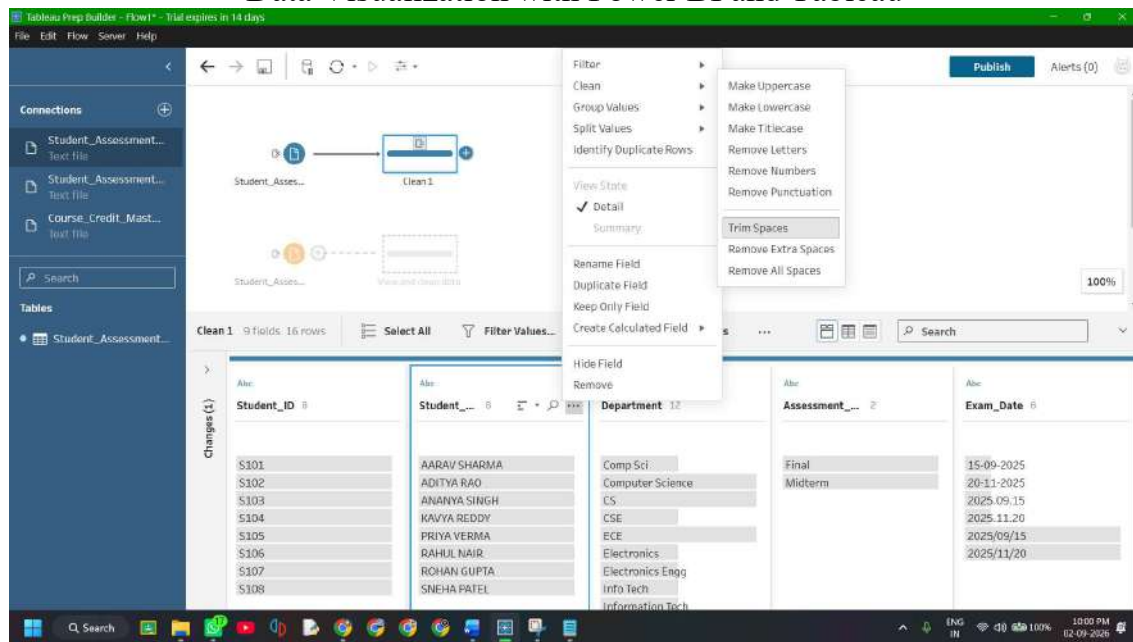
1. Click on the **Clean 1** step connected to `Student_Assessments_Term1`.
2. Look at the cards displayed in the **Profile Pane**:
 - Each card represents one column.
 - Notice that **Student_Name** has multiple representations of the same student due to extra spaces and mixed casing (e.g., ' Aarav Sharma ' vs 'Aarav Sharma', 'priya VERMA', 'aditya Rao ').
 - Notice that **Exam_Date** is recognized as a string (Abc) due to mixed separators (/, -, .).



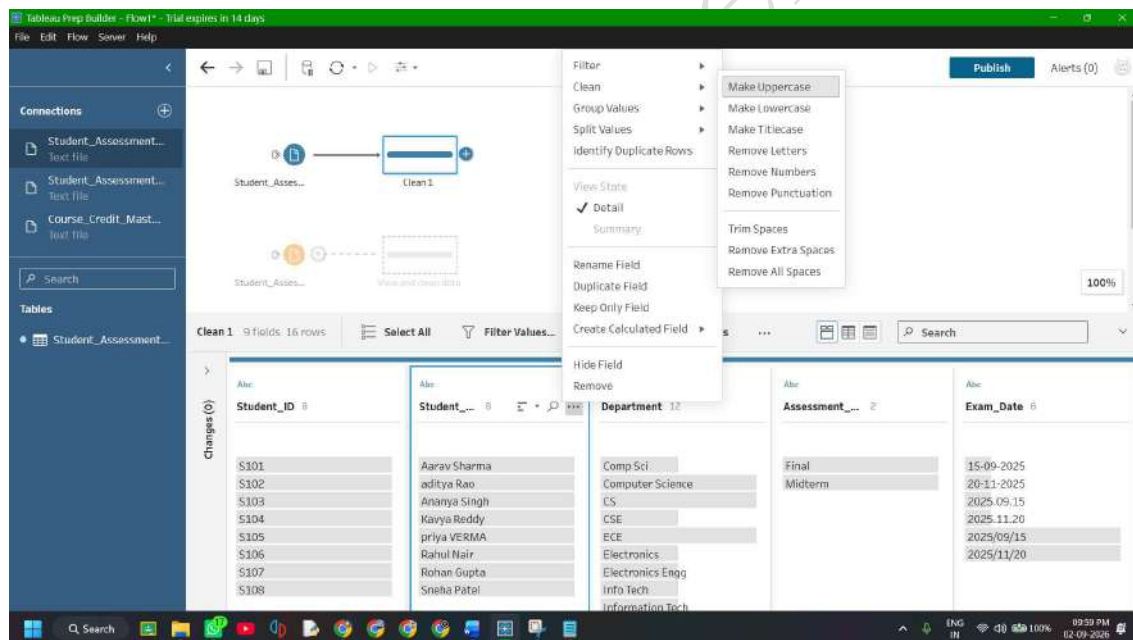
Step 2: Clean Whitespace and Standardize Letter Casing

1. On the **Student_Name** profile card:
 - Click the **More options (...)** icon at the top right of the card.
 - Go to **Clean** --> select **Trim Spaces** (this strips all leading and trailing whitespace).

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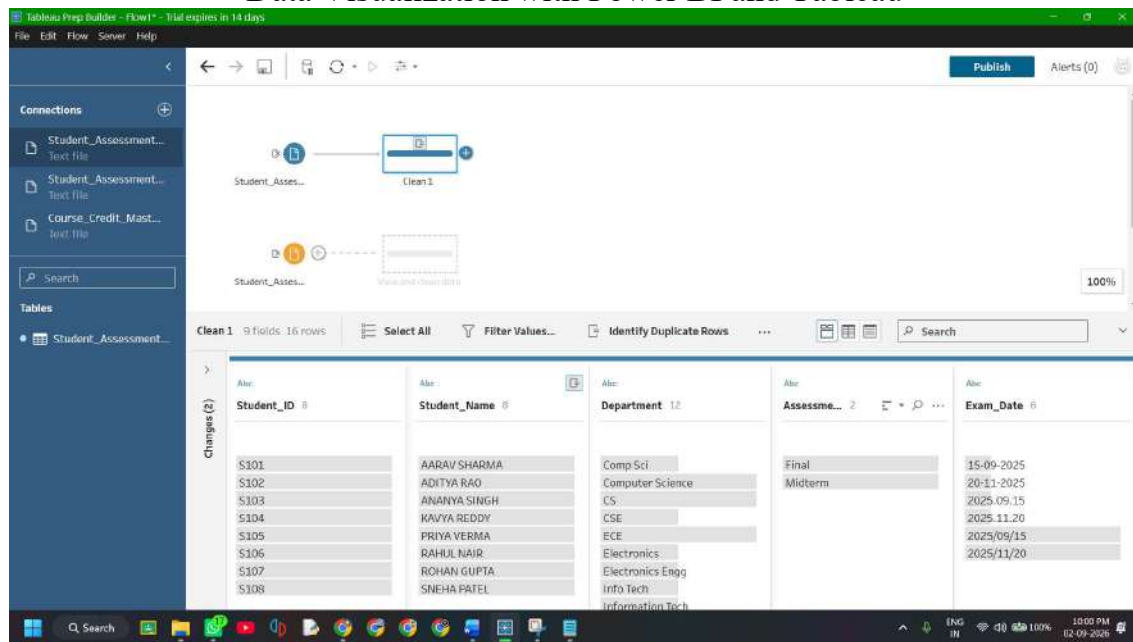


- Click **More options (...)** again --> **Clean** --> select **Make Uppercase** (or **Make Lowercase** / standard proper formatting).



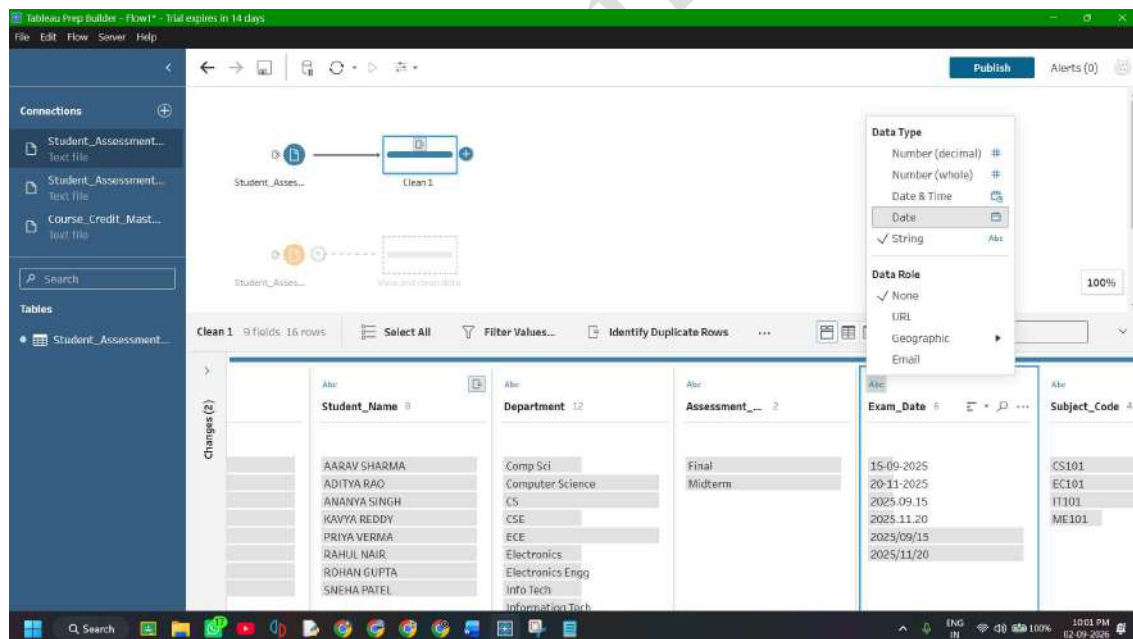
2. The card updates immediately, consolidating duplicate whitespace variations.

Data Visualization with Power BI and Tableau



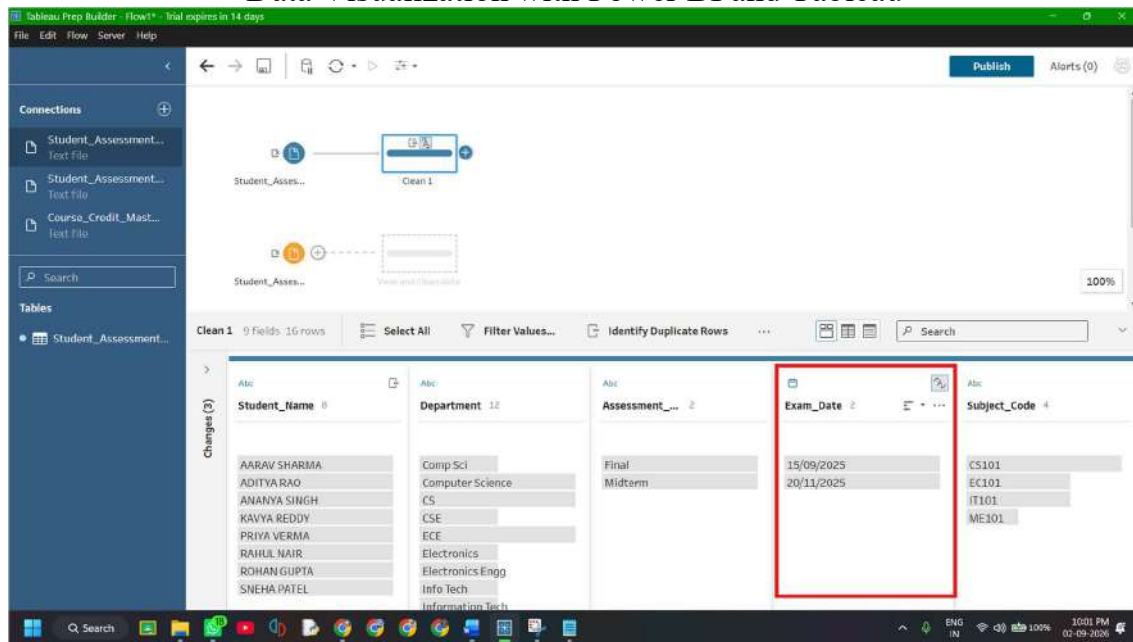
Step 3: Convert Exam_Date to Date Data Type

1. Locate the **Exam_Date** profile card.
2. Click the data type icon (currently showing **Abc**) directly above the column name.
3. Select **Date** from the dropdown menu.



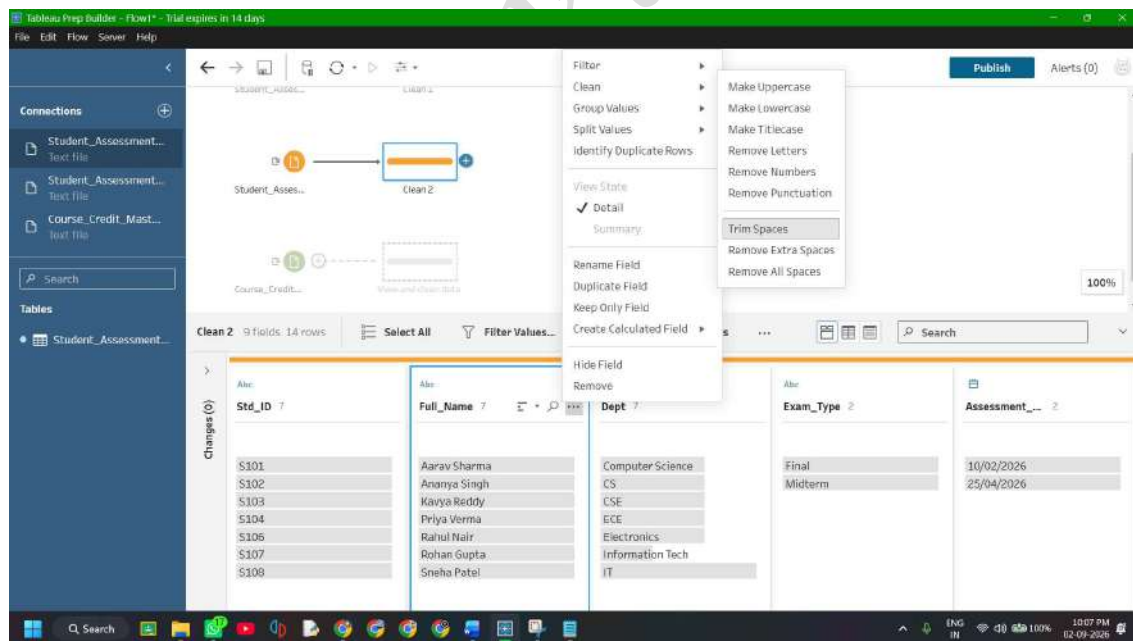
4. Tableau Prep automatically parses the various date formats (2025/09/15, 15-09-2025, 2025.11.20) into standardized calendar dates.

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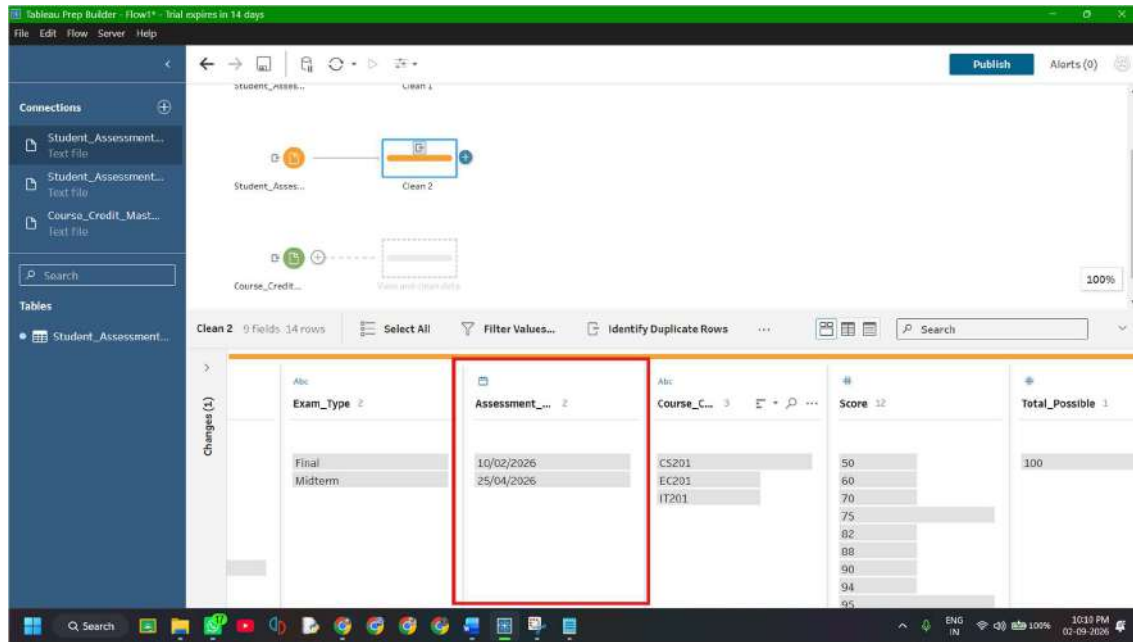
Step 4: Clean the Second Dataset (Student_Assessments_Term2.csv)

1. On the Flow canvas, hover over the **Student_Assessments_Term2** input icon.
2. Click + --> select **Clean Step** (creates **Clean 2**).
3. On the **Full Name** card in Clean 2:
 - Click **More options (...)** --> **Clean** --> **Trim Spaces**.



4. On the **Assessment_Date** card:
 - Click the **abc** icon --> change the data type to **Date**.

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Note: Tableau Prep may have already Selected the date datatype.

Step 5: Review the Changes Pane

1. Look at the left side of the screen under the **Changes** tab.
2. Notice that Tableau Prep records every single transformation as an itemized, reversible step (e.g., *Trim spaces on Student_Name*, *Change data type of Exam_Date*).

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Profile & Cleaning Audit Table

Field Name	Detected Issue	Cleaning Operation Applied	Target State
Student_Name	Leading/trailing spaces (' Aarav Sharma ')	Clean → Trim Spaces	Single trim-aligned name
Student_Name	Inconsistent casing ('priya VERMA')	Clean → Make Uppercase	Standardized casing
Exam_Date	Text datatype with mixed separators	Change Data Type → Date	Standard Date format (YYYY-MM-DD)
Assessment_Date	String data type	Change Data Type → Date	Standard Date format (YYYY-MM-DD)